

Unit

2

Lesson

1



Innovators in Action

Building a Talking Machine.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Demonstrate how adding components (like LEDs) in series affects the circuit.
- Applying operations (addition, multiplication, division) to solve real-world problems.
- Apply knowledge of series circuits by building a project on a smartpad.
- Use a push button to control when the recorded sound plays.
- Connect a battery, DFPlayer module, and speaker correctly to create a working circuit.

Challenge questions of the lesson

1. What happens to a circuit when we add more components like LEDs?
2. How can we use math to figure out how much power our circuit needs?
3. Can we build a smart circuit that only plays a sound when we press a button?
4. What is the right way to connect a battery, DFPlayer, and speaker to make a sound?
5. What happens if one part of our sound machine is missing or not connected properly?

SEL Relation

- Responsible Decision-Making: Designing a circuit that solves a real-world challenge by giving someone a voice.
- Self-Management: Persisting through trial and error while building and testing a sound-playing machine.
- Relationship Skills: Working together to share ideas, divide tasks, and support teammates during circuit construction.
- Social Awareness: Understanding how assistive technology can improve communication for people with different abilities.
- Self-Awareness: Recognizing how their actions can help others feel included, understood, and heard.

Engage



Teacher's Role:

Introduce the Activity

- Set the mood by dimming the lights or playing a short audio clip of a power outage (e.g., thunder, lights flickering).
- Ask the class:
 - "Have you ever experienced a power outage? What happened when the electricity went out?"
 - "What things stopped working, and how did it feel?"

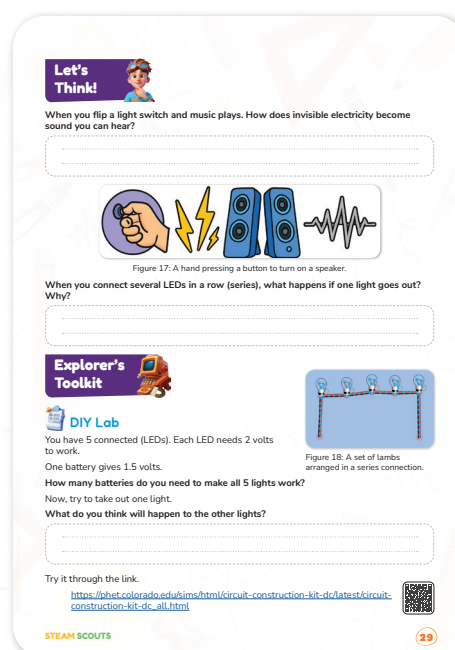
Facilitate a Class Discussion

- Lead a whole-class discussion with open-ended questions:
 - "What do you think happens when you flip a light switch?"
 - "How do you think electricity helps a speaker make sound?"
- Encourage students to turn and talk to a partner before sharing aloud.
- Reinforce that all ideas are welcome — this is a space to wonder and explore.

Expected From Students:

- Watch the animated story video attentively.
- Discuss their ideas in groups.
- Share their thoughts with the class

Parts of the books included.



Story



1. Gather the class and set a collaborative tone.
2. Say: "STEAM Scouts, we've explored communication and how sound works, especially thinking about how to help our friend Omar. Now, we're ready to build! To make a talking machine, we need to understand the invisible power that makes devices work: electricity!"
3. Read the story aloud using expression and pacing to emphasize curiosity and teamwork. (Read the following story scenario:)

"Ms. Sara challenged everyone: 'Remember how we talked about using science to solve problems? Omar needs our help!' Adam quickly suggested, 'What if we could build a machine that could say words for Omar when he types them?' Omar wrote on his notepad: 'Yes! But it needs to be fast and easy.' Ms. Sara smiled, 'Omar has given us a wonderful challenge! How can we build a machine that's not just a talking machine, but one that's also fast and easy for Omar to use?'"

4. Prompt students with guiding questions:

- "What did Adam and Lila suggest?"
- "Why do you think Omar wants the machine to be fast and easy?"
- "What kind of help does Omar need from his friends?"

Possible Strategies for Implementation:

- **Create an Immersive Atmosphere:**
 - Set the Scene: Begin by reminding students of the previous lesson's context (e.g., Omar's communication challenge, the idea of a talking machine).
 - Visual Cues: If possible, have a relevant image or a simple drawing on the board that sets the mood (e.g., a simple sketch of a robot or circuit, or even a picture of Stephen Hawking if you referenced him in the previous lesson).
- **Engage with Expression and Pacing:**
 - Vary Voice: Read the story with enthusiasm, changing your voice for different characters (Ms. Sara, Adam, Omar) to make it more dramatic and captivating.
 - Strategic Pauses: Pause after key lines or questions to allow students to absorb the information and build anticipation.
 - Body Language: Use gestures and facial expressions to convey excitement and curiosity.
- **Encourage Active Listening:**
 - "Listen for...": Before reading, tell students to listen for specific things, such as "Listen for what Adam suggests," or "Listen for what Omar's challenge is."

- **Facilitate Discussion with Guiding Questions:**

- Think-Pair-Share: After asking the guiding questions (e.g., "What did Adam and Lila suggest?"), give students a moment to think individually, then discuss with a partner, and finally share with the whole class.
- Open-Ended Exploration: Frame questions to encourage multiple perspectives and avoid "yes/no" answers. Validate all thoughtful responses.
- Connect to Personal Experience: Ask if students have ever had difficulty communicating or seen someone else struggle, to build empathy.

Let's Think!



Objective:

To activate prior knowledge about electricity and sound, introduce the concept of circuits (especially series circuits), and spark curiosity before hands-on building.

Answers:

How does invisible electricity turn into sound/light?

- Electricity flows through a circuit to a device (like a speaker or light bulb).
- Inside the speaker, electricity causes parts (like a cone) to vibrate, which creates sound waves.
- Inside a light bulb, electricity heats a filament or excites gases, producing light and sometimes a little heat.
- The key is that electricity is a form of energy that can be transformed into other forms of energy, like sound and light, by different components.

What happens if one light goes out in a series circuit? Why?

- If one light goes out (or breaks) in a series circuit, all the other lights will also go out.
- Why: In a series circuit, electricity has only one path to flow through. If any component in that path breaks or is removed, the circuit becomes "open," and the flow of electricity is interrupted for the entire circuit. No electricity can reach the other lights, so they all turn off. This illustrates the concept of a complete vs. incomplete circuit.

What might that tell us about how the electricity is flowing through these lights?

- It tells us that the electricity is flowing in a single, continuous loop, and if that loop is broken anywhere, the flow stops for everything connected in that loop. This is the defining characteristic of a series circuit.
- **Facilitate Discussion:** Encourage students to share their initial thoughts and predictions. Guide them to connect the idea of an unbroken path (circuit) for electricity to flow. Emphasize that there are no wrong answers at this stage – it's about exploring ideas!

Possible Strategies for Implementation:

- **Brainstorming and Prior Knowledge Activation:**
 - Open Discussion: Allow students to freely share their initial thoughts and ideas about how electricity becomes sound/light. Write their ideas on the board without judgment.
 - Connect to Daily Life: Ask students to think about everyday examples where electricity powers sound (e.g., radio, TV, phone calls).
- **Hands-On Observation (if applicable):**
 - Simple Demonstrations: If feasible and safe, have a simple string of holiday lights (like in the image) and demonstrate what happens when one bulb is unscrewed. This immediate visual feedback can be powerful.
- **Guided Inquiry:**
 - "What do you observe?": For the light string example, ask them to describe exactly what they see happening.
 - "What does that tell us?": Guide them to infer the implications of their observations (e.g., "If one goes out and they all go out, what does that mean about how they're connected?").
 - Avoid Giving Answers Too Soon: Reinforce that this is a "think" stage, and they are exploring ideas. The goal is to generate hypotheses, not necessarily to get the "right" answer immediately. The actual answers will be revealed in the upcoming "DIY Lab" and "Screen Lessons."
- **Visual Reinforcement:**
 - Refer to the Image: Constantly refer back to the "Let's Think" image (3.PNG) as you pose the questions, helping visual learners connect the concepts.
- **Encourage Collaboration:**
 - Small Group Discussion: Have students discuss the "Let's Think" questions in small groups before sharing with the whole class. This can build confidence and allow more students to participate.

Explorer's Toolkit



Teacher's Role:

- Introduce the engineering challenge by asking a real-world question about powering multiple LEDs using batteries.
- Ask guiding questions.
- Encourage students to estimate and write their predictions on paper.
- Support students in using addition or multiplication to calculate total voltage needs.
- Model the experiment live (or use a PhET simulator) by building a series circuit with 1 LED, then gradually increasing to 5 LEDs.

- Pause at each step to prompt discussion.
- Let students test their ideas in pairs using tablets or laptops.
- Circulate and support their thinking with questions like:
 - “What might be stopping the lights from working?”
 - “Can you try using more batteries?”
 - “What happens if you remove one LED from the circuit?”
- Facilitate a wrap-up discussion where groups share their findings.
- Reinforce that engineers test, revise, and learn from what doesn’t work, too.

Expected From Students:

- Listen carefully to the challenge and think critically about the problem.
- Predict how many volts are needed to power 5 LEDs and how many batteries are required.
- Use math strategies like addition or multiplication to estimate total voltage.
- Observe carefully as the teacher models a circuit, noticing how brightness changes with more LEDs.
- Work in pairs to test circuits using the simulator or physical components.
- Try different combinations of batteries and LEDs to explore what works.
- Pay attention to what changes when an LED is removed from a series circuit.
- Discuss and share findings with the class, including surprises or patterns they noticed.
- Reflect on what worked, what didn’t, and how engineers adjust and improve their ideas.
- Show persistence and teamwork while solving problems through trial and error.

Parts of the books included.



Explorer's Toolkit



DIY Lab

You have 5 connected (LEDs). Each LED needs 2 volts to work.
 One battery gives 1.5 volts.
How many batteries do you need to make all 5 lights work?
 Now, try to take out one light.
What do you think will happen to the other lights?

Try it through the link.
https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_all.html



STEAM SCOUTS

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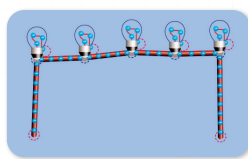


Figure 18: A set of lamps arranged in a series connection.



Objective:

Students will explore how electricity flows in a series circuit, use math to calculate voltage requirements, and predict what happens when one component is removed. This introduces them to the concept of electrical current distribution in real-world systems like assistive technology.

Link to Activity: https://phet.colorado.edu/sims/html/circuit-construction-kit-dc/latest/circuit-construction-kit-dc_all.html



Answer:

How many batteries do you need to make all 5 lights work?

Each LED needs 2 volts, and there are 5 LEDs, so:

$5 \text{ LEDs} \times 2 \text{ volts} = 10 \text{ volts needed}$

Each battery gives 1.5 volts, so:

$10 \div 1.5 = 6.67 \text{ batteries} \rightarrow \text{You need at least 7 batteries to power all 5 LEDs.}$

What do you think will happen if you take out one light in a series circuit?

All the other lights will go off.

In a series circuit, if one light is removed or stops working, the whole circuit breaks and no electricity flows to the other lights.

Activity Instructions for Teachers:

1. Introduce the Scenario:

- Say: “You have 5 connected lights (LEDs). Each need 2 volts to work. One battery gives 1.5 volts.”
- Ask: “How many batteries do you think are needed to light up all 5?”

2. Challenge Students to Calculate:

- Encourage them to estimate using addition or multiplication:
 - $5 \text{ LEDs} \times 2 \text{ volts} = 10 \text{ volts}$
 - $10 \div 1.5 = \sim 6.7 \rightarrow \text{They'll need 7 batteries!}$

3. Use the Simulator:

- Share the PhET link and guide students to build a series circuit using 5 bulbs and multiple batteries.
- Test different combinations to see what works.

4. Explore the Concept of Interruption:

- Ask them to remove one light in the series and observe what happens to the rest.
- Ask: “What does this teach us about how energy flows?”

5. Discuss Results:

- Facilitate a group discussion on their findings, surprises, and real-world connections (like why lights sometimes go out together!).

Possible Strategies for Implementation:

- **Math Integration:**
 - Use real-world voltage calculations to connect math and science.
- **Predict-Test-Reflect Cycle:**
 - Have students write down predictions before building, then reflect after testing.
- **Visual Learning:**
 - Display the simulator on a projector as a class model before individual work.
- **Collaborative Work:**
 - Pair students to build, problem-solve, and explain their circuits together.
- **Guiding Questions:**
 - “What happens when you add more batteries?”
 - “Why do all the lights go off when one bulb is removed?”

Explain



Teacher's Role:

- Introduce the video by saying:
- “Let’s watch a short video to understand how series circuits work and how electricity flows through them.”
- Set the purpose before playing:
- “While you watch, think about: What happens when we add more bulbs? What if one bulb is removed?”
- Pause at key points and ask guiding questions like:
 - “What do you notice about brightness?”
 - “Is the electricity shared or separate in a series circuit?”
- Lead a short discussion after the video to clarify ideas and connect to real-world circuits (like holiday lights or toys).

Expected from Students:

- Watch the video attentively.
- Observe how electricity flows through a series circuit.
- Think about what happens when:
 - More bulbs are added.

- One bulb is removed or breaks.
- Participate in discussion by sharing what they observed or learned.
- Begin to apply what they saw to their simulation or real circuits in the DIY Lab.

Parts of the books included.

Screen Lessons

Watch the video and learn more about series circuits.

<https://youtu.be/js7Q-r7G9ug?si=cTjQNmSSwfK-zPTK> (till min 5)

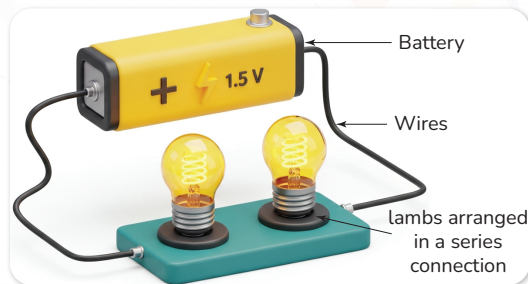


Figure 19: Electric circuit.

Screen Lessons

Objective:

Students will understand how electricity flows through a series circuit and how the number of components (like bulbs) affects brightness and functionality. They will explore the concept of shared energy and circuit dependency.

Video Link: <https://youtu.be/js7Q-r7G9ug?si=cTjQNmSSwfK-zPTK>
(Watch until minute 5)

Possible Strategies for Implementation:

1. Pre-Watching

- Set a Purpose for Watching
 - Explain what students should look for in the video (e.g., how electricity flows, what happens when part of the circuit is broken).
 - “As you watch, think about why all the lights go out when one part of the circuit is removed.”
- Activate Prior Knowledge
 - Review what students observed in the simulator. Ask: “What happened in your test when you removed one LED?”

2. During Watching

- Pause for Key Questions
 - Stop at key points in the video (e.g., when the circuit is connected, or when it breaks) and ask: “Why do the lights go out here?”

“What does this tell us about how series circuits work?”

- Highlight Vocabulary
 - Point out and clarify terms like flow, resistance, complete path, and interruption when they appear in the video.
- Use Visual Anchoring
 - Draw or display a simple circuit diagram on the board to help students track the concept being explained.

3. Post-Watching

- Think-Pair-Share
 - Ask students to explain what they learned to a partner, then share with the class.
- Class Discussion
 - Prompt deeper thinking with questions like:
“What surprised you?”
“Does this explain what happened in your simulation?”
- Quick Reflection
 - Have students complete a prompt:
“One thing I learned about series circuits is...”
or
“Now I understand why...”

Elaborate



Teacher's Role

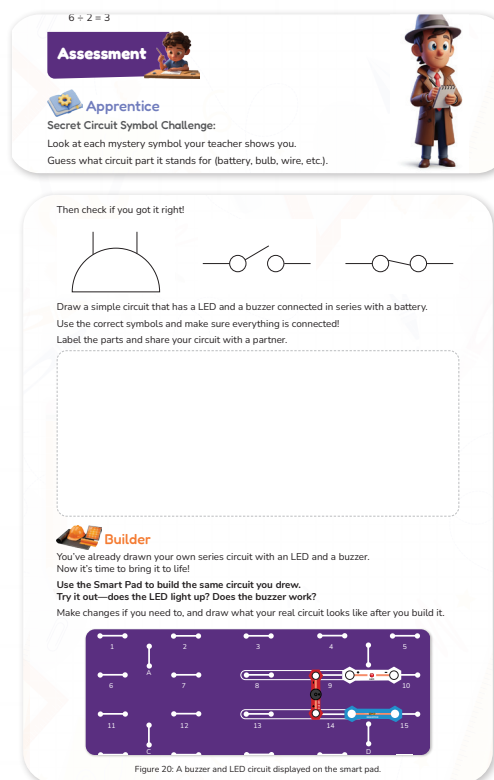
- Introduce the “Secret Circuit Symbol Challenge” by showing students visual cards of circuit symbols.
- Guide students in identifying common symbols (battery, wire, bulb, buzzer) and labeling them correctly.
- Support students as they draw their own circuits using correct symbols with one LED and one buzzer in series.
- Monitor for symbol accuracy and ensure components are logically connected.
- Transition to the Builder activity by helping students match their drawn circuits to actual Smart Pad setups.
- Provide materials and demonstrate how to safely use the Smart Pad.
- Encourage students to test and troubleshoot their circuit: “Does the LED light up? Can you hear the buzzer?”
- Prompt design revision if needed, and ask students to redraw the real-life circuit after testing.

- Celebrate all circuit completions and emphasize learning from mistakes.

Expected From Students

- Accurately identify and match electrical symbols (battery, LED, buzzer, wire).
- Draw a basic circuit using symbols, showing correct connections.
- Build their drawn circuit using the Smart Pad and test its function.
- Revise their circuit if something doesn't work, and draw the updated version.
- Work in pairs or independently to solve problems and check connections.
- Share their working circuit and explain how it functions.
- Show persistence and curiosity when things need fixing.

Parts of the books included.



Apprentice

Objective:

- Students will demonstrate their understanding of circuit components by identifying standard symbols.
- Students will apply their knowledge of series circuits by drawing and labeling a complete, functioning circuit using correct symbols and connections.

Resources: <https://docs.google.com/presentation/d/1vYrka3DSj8xvOaeEVwNpQcx8EbKnWsyL/edit?slide=id.p1#slide=id.p1>

Activity instructions:

Part 1: Secret Circuit Symbol Challenge

1. Prepare 7–5 circuit symbols on flashcards or slides" use PowerPoint " (battery, LED,



buzzer, switch, wire, etc.).

2. Present each symbol one at a time to the class.
3. Ask:
 - “What do you think this symbol represents?”
 - “Where might you see this in a real circuit?”
4. After guesses, reveal the correct answer and briefly explain its function in a circuit.

Part 2: Circuit Drawing

1. Give students paper or a worksheet with space to draw “They can draw in the book”.
2. Prompt:
 - “Draw a series circuit with a battery, an LED, and a buzzer. Use the correct circuit symbols and label each part.”
3. Circulate while students draw, offering hints or asking clarifying questions:
 - “Is the circuit complete?”
 - “Does your drawing show how electricity flows?”
4. Have students share their drawing with a partner and explain how it works. Encourage them to check if each component is connected properly.

Possible Strategies for Implementation

- **Use Visual Aids.**
 - Display clear, familiar circuit symbols on the board or distribute a symbol reference chart to support recognition and recall during the challenge.
- **Game-Based Engagement.**
 - Turn the symbol challenge into an interactive quiz or game. Use flashcards, slides, or quick-response activities (e.g., mini whiteboards or finger voting) to boost participation and focus.
- **Model Drawing Expectations.**
 - Demonstrate how to draw a basic series circuit step-by-step. Verbalize your thought process to show how each component fits and why accurate connections matter.
- **Think-Pair-Share.**
 - Encourage students to compare and discuss their circuit drawings with a partner before sharing with the class. This builds communication and reinforces understanding.
- **Peer Feedback.**
 - Guide students to check each other’s work using a simple checklist:
 1. Are all parts labeled?
 2. Is the circuit complete?

3. Are symbols used correctly?

- **Scaffold the Task.**

- Provide partially completed diagrams for students who need support, or allow advanced students to extend their drawings with extra components (e.g., switches or resistors).
- Reflection Exit Slip
- Wrap up the activity by asking students to write one thing they learned and one thing they still wonder about circuits or symbols.



Builder

Objective:

- Students will construct a functioning series circuit using the Smart Pad and electronic components.
- They will compare their physical build to their original drawing and revise their diagram based on real-life results.
- Students will apply problem-solving and iterative design skills by testing, improving, and documenting their circuit.

Activity Instructions:

- **Review Your Sketch.**

- Take out the series circuit drawing you created in the Apprentice activity.
- Double-check that it includes a battery, an LED, and a buzzer—all connected in a single line (series).

- **Gather Materials.**

- Smart Pad (circuit board)
- Components: battery holder, wires, LED, buzzer

- **Build the Circuit.**

- Place Components on Magnetic Connectors:

1. The Smart Pad uses magnetic connection points. Each component (battery, wires, LED, buzzer) is held in place using these magnetic terminals.
2. Students should gently place and align each piece on the pad according to their circuit sketch.

- **Polarity Matters!.**

- LEDs and buzzers have a positive (+) and negative (–) side.
- The longer leg of the LED is the positive (anode) and should connect to the positive terminal of the battery.
- The shorter leg (negative/cathode) connects toward the buzzer's positive leg, forming a continuous path.

- **Correct Sequence of Components:**

- The typical order should be: Battery (+) → LED → Buzzer → Battery (-).
- This ensures that current flows in one direction and both the LED and buzzer work properly.

Make sure the power is off while you're connecting components.

- **Test Your Circuit**

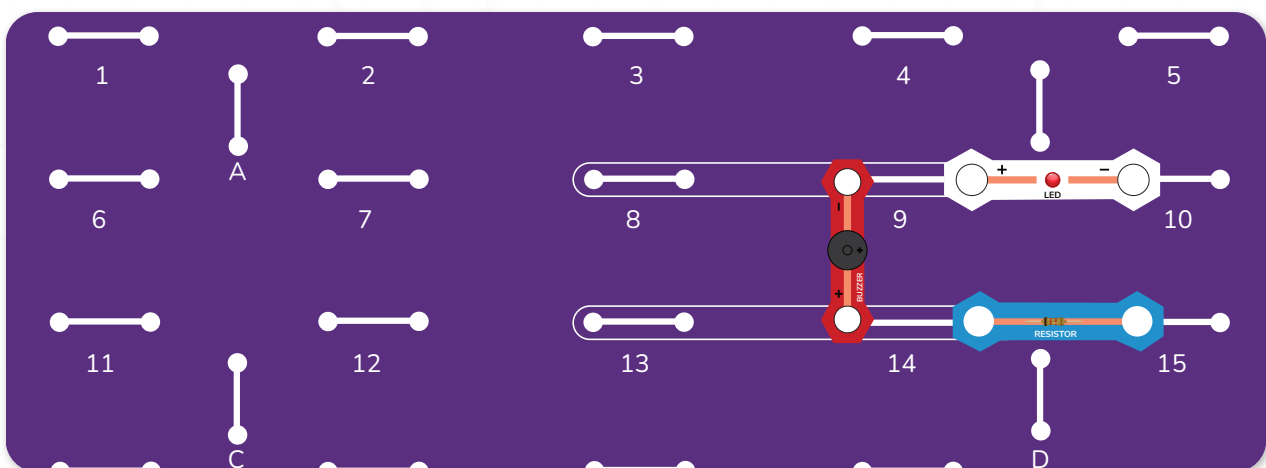
- Turn on the Smart Pad.
- Observe: Does the LED light up? Do you hear the buzzer?

- **Troubleshoot If Needed**

- If it doesn't work, check connections or try changing the order.
- Ask yourself: "Is everything connected in one continuous loop?"

- **Reflect and Record**

- On the same paper or new space, draw what your final working circuit looks like.
- Label the components clearly.



Evaluate



Teacher's Role:

- Guide students as they review their circuit sketches and compare them with the actual breadboard layout.
- Provide labeled pictures or real components (speaker, buttons, breadboard, Arduino) for matching and identification.
- Support students as they place the buttons and speaker in the correct positions on the breadboard.
- Check wire connections, ensuring students understand why each component goes where it does.
- Ask prompting questions like:

- “What happens when we press the button?”
- “Why is the speaker connected to the Arduino and not directly to the battery?”
- Celebrate student progress and build excitement for testing the full voice circuit in the next lesson.
- Emphasize safety: make sure power is off while assembling.

Expected from Students:

- Carefully examine the breadboard layout and identify each component: button, speaker, and wires.
- Match each physical part to their sketch or worksheet.
- Add the final components to the breadboard (with teacher support).
- Discuss as a group how the system will work once fully connected.
- Reflect on how their device could help someone who cannot speak.
- Ask questions and make suggestions for final tweaks to the prototype.



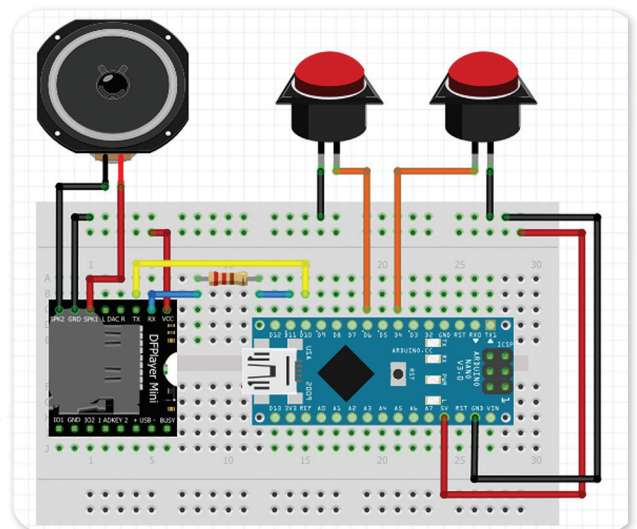
Showcase

Objective:

To complete the communication device prototype by adding a pushbutton and a speaker to the breadboard, simulating how a real talking device for people with speech difficulties could be built.

Activity Instructions :

1. Review the Circuit So Far:
 - Ask students to look at their previous circuit (with Arduino, LED, and buzzer).
 - Discuss what still needs to be added to make it work like a real communication device.
2. Introduce the New Components:
 - Show students the button and speaker.
 - Explain: “The button sends a signal to tell the speaker when to talk.”
 - Point out how the speaker plays sound, just like our voice.
3. Help Students Match the Parts:
 - Give students a labeled diagram or the image on the page.
 - Ask them to circle or label the button and speaker.
4. Support Assembly (teacher-led for safety):
 - Walk students through placing the button and speaker on the breadboard.



- Ensure they check the wiring is connected properly to the Arduino and power source.
5. Discussion Prompt:
- Ask: “How is this circuit helping someone like Omar?”
 - “What do you think this button will do when it’s fully programmed?”

Step for teachers to connect the circuit.

Connections

DFPlayer Mini → Arduino Nano (or uno according to what available in school)

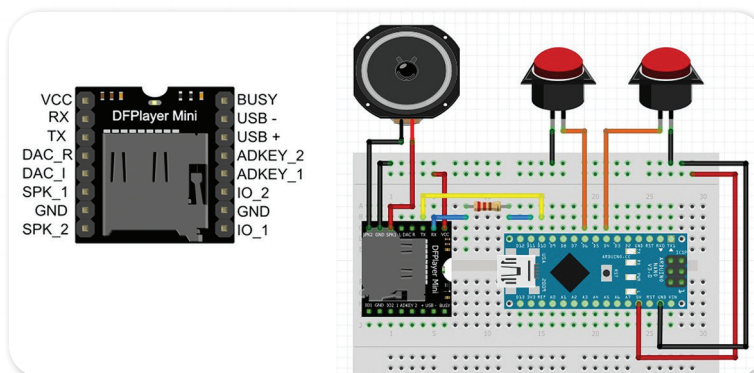
DFPlayer Mini Pin	Connect To
VCC	Arduino 5V
GND	Arduino GND
TX	Arduino D10
RX	Arduino D11 (via 1kΩ resistor)

Speaker ↔ DFPlayer Mini

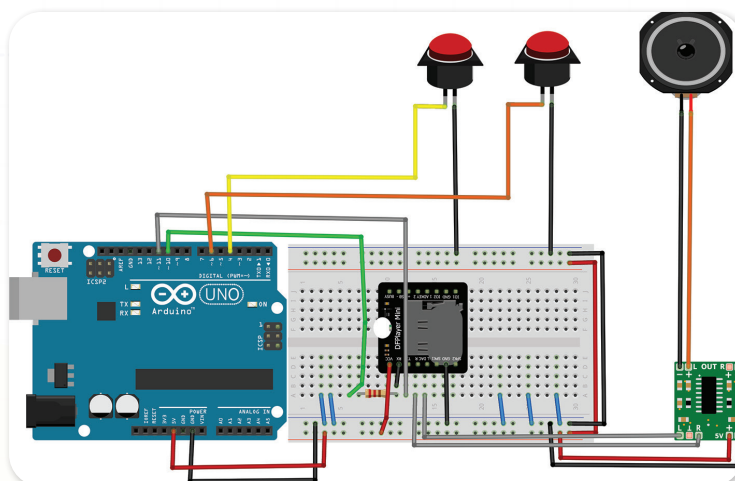
DFPlayer Mini Pin	Connect To
SPK1	Speaker +
SPK2	Speaker –

Buttons ↔ Arduino

Push Button	Connects To
Button 1 (Track 1)	One leg to D4 Other leg to GND
Button 2 (Track 2)	One leg to D6 Other leg to GND



Note: If you faced any problem making the circuit work, connect amplifier as following.



Possible Strategies for Implementation:

1. Model Before Making

- Project the sample circuit diagram on a large screen.
- Walk through each part (Arduino, speaker, button) and ask students to identify the components on their own breadboards. “But no need to go deeper in Arduino and breadboard, just mention the Arduino is the brain of the robot and bread board to add our components on.”

2. Pair and Share Assembly

- Let students work in pairs: one places components, the other checks connections.
- Switch roles to give each learner hands-on experience.

3. Use Color-Coded Wires

- Provide pre-sorted wire colors (e.g., red for power, black for ground, yellow for signal) to make wiring clearer and reduce confusion.

4. Error Exploration

- If the speaker doesn’t work, guide students to debug by checking each component.
- Ask: “Did you press the button?” “Is the speaker in the right pins?”

5. Peer Teaching Corners

- Assign confident students to “tech support” corners where others can go for help before asking the teacher.

6. Explain Your Build

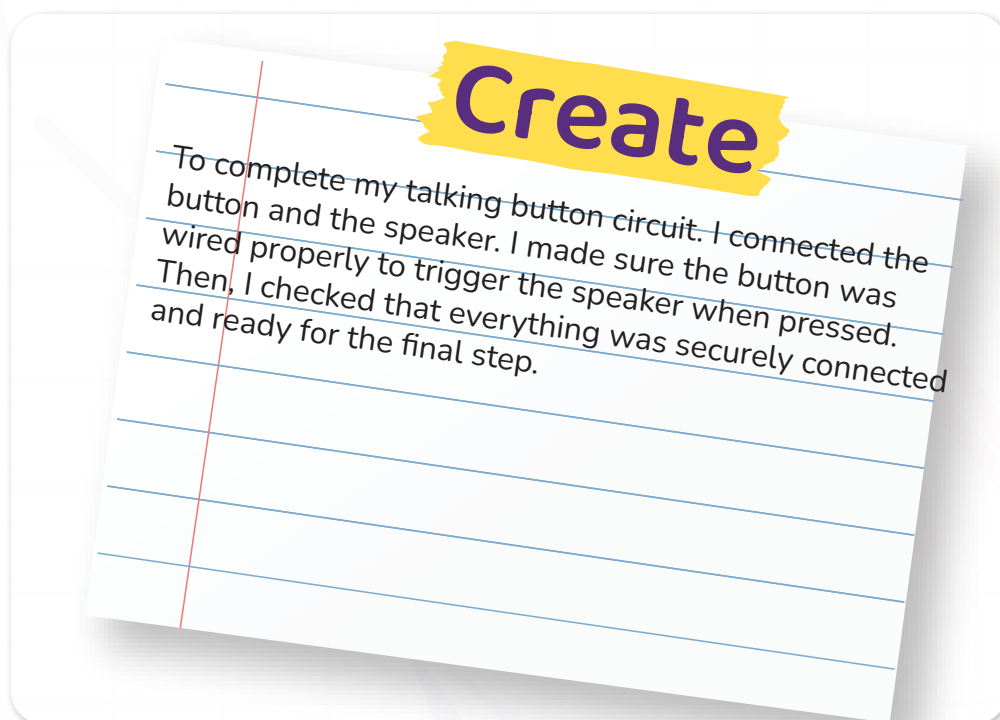
- After building, each student or group presents how their circuit works and what it helps solve.

Showcase: Rubric for Evaluating circuit completion activity.

Criteria	Excellent (25 pts)	Good (20 pts)	Developing (15 pts)	Needs Improvement (10 pts)
Circuit Accuracy	All components are correctly placed and wired.	One minor error in wiring, but circuit works.	Multiple wiring issues; needs assistance to function.	Major errors; circuit incomplete or incorrect.
Understanding Purpose	Clearly explains how the button and speaker simulate real communication devices.	Explains the general idea with minor gaps.	Gives a basic explanation with support.	Cannot explain or connect function to purpose.

Criteria	Excellent (25 pts)	Good (20 pts)	Developing (15 pts)	Needs Improvement (10 pts)
Teamwork & Communication	Collaborates fully, shares roles, and discusses ideas respectfully.	Works well with minor teacher prompting.	Occasionally collaborates, needs reminders.	Rarely collaborates or dominates the task.
Presentation & Labeling	Diagram is neatly labeled, circuit matches drawing.	Diagram mostly clear, some minor label issues.	Basic drawing, few labels.	Unclear or missing drawing and labels.

Don't forget to go back to the EDP page and add these to planning part.





Now I Can

Objective:

To help students self-assess their understanding of series circuits and how circuits can be used to build assistive technology (e.g., a talking button system). This reflection ties together science, engineering, and real-world application

Activity Instructions (for Teachers):

Step 1 – Read Aloud Together.

- Project or read each "Now I Can" statement aloud. Clarify any tricky words like “division,” “circuit,” or “speaker.”

Step 2 – Think-Pair-Share.

- Ask students to reflect individually first: “Which of these did you do today?”
- Then pair up and have them share examples: “How did you connect a battery to a speaker?”

Step 3 – Student Checkmarks.

- Let each student check off the boxes that match what they did or learned.
- Encourage honesty—this is about learning growth, not perfection.

Step 4 – Optional Extension.

- Ask students to draw one step from the circuit or write 1–2 sentences on the back:
 - “The hardest part was...”
 - “I want to learn more about...”

Step 5 – Wrap-Up Discussion.

- Close with a group talk:
 - What surprised you today?
 - How could a talking circuit help someone in real life?

Relation to Standard

1. NGSS (Next Generation Science Standards)

3-PS2-4: Define a simple design problem that can be solved by applying scientific ideas about magnets or electric circuits.

3-5-ETS1-1: Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.

3-PS2-3: Ask questions to determine cause-and-effect relationships of electric or magnetic interactions.

3-PS2-2: Make observations and/or measurements of an object’s motion to provide evidence that a pattern can be used to predict future motion (linked to understanding how electricity flows in a circuit pattern).

2. ISTE (International Society for Technology in Education)

1.4 Innovative Designer: Students use a deliberate design process to generate ideas, test theories, create innovative artifacts, or solve authentic problems.

1.6 Creative Communicator: Students choose tools to demonstrate their learning in varied formats (e.g., building circuits, sketching, speaking, etc.).